Slopes and local change

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For one example with prediction 4 and target 2, what is the squared error?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Calculate an average rate of change over a stated interval. Try the recall before opening notes.

Your learning target

I can calculate an average rate of change over a stated interval.

I can estimate a local **slope** with nearby loss values.

I can explain the limits of a local direction clue.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Calculate an average rate of change over a stated interval

Estimate a local **slope** with nearby loss values

Explain the limits of a local direction clue

Retrieve what you already know

> For one example with **prediction** 4 and target 2, what is the squared error?

My starting explanation

This week's end goal

By the end, I can explain **slope**s and **local change** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Slopes and local change case. Separate observations, given facts and assumptions before choosing an explanation.

Let a one-example model predict w and let its target be 2. Its squared loss is L(w) = (w - 2) squared. Here w is the parameter we vary, not a new data example. L is a nonnegative function of that parameter.

w	Prediction minus target	L(w)
0	-2	4
1	-1	1
2	0	0
3	1	1
4	2	4

From w = 0 to w = 1, the loss changes by -3 while w changes by 1. The average **slope** is -3. From w = 1 to w = 2 it is -1. The curve does not have one fixed **slope** everywhere.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Slopes and local change. Connect each labelled input, deciding step and result.

Near w = 1, use two symmetric nearby values: 0.9 and 1.1. Their losses are 1.21 and 0.81. Change in loss divided by change in w gives (0.81 - 1.21) / (1.1 - 0.9) = -2.

This symmetric estimate describes change near the centre w = 1.

```
def loss(w):
    return (w - 2) ** 2

def local_slope(w, step):
    if step <= 0:
        raise ValueError("Step must be positive")
    return (loss(w + step) - loss(w - step)) / (2 * step)

for w in [1, 2, 3]:
    print(w, round(local_slope(w, 0.1), 6))
# Slopes near these centres: -2, 0, 2
```

For this quadratic, the symmetric estimate equals the exact **slope** 2(w - 2) in exact arithmetic. Numerical estimates for other functions are generally approximate.

Smaller steps can reduce some approximation errors, but extremely small steps can suffer from computer rounding.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Slopes and local change. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Move from 0 all the way to5.

Predicted result

Prediction: Loss becomes9, exceeding the starting4.

Why the result follows

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Words that unlock the idea

> Use the vocabulary of Slopes and local change to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Slope	Change in output divided by change in input over a stated interval
Local change	Change considered near a particular input or parameter value
Step size	The amount by which an input or parameter is changed
Stationary point	A point where a differentiable function has zero local slope

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Slopes and local change, predict a step, then compare with its explanation.

Compare slope clues on the same loss curve

CURRENT W	LOCAL SLOPE	NEARBY DOWNHILL DIRECTION
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

At w = 3, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Slopes and local change

Use a local **slope** to propose a small move rather than claiming every move helps.

The facts we will use

Supplied facts

Toy loss $J(a)=(a-2)^2$. At $a = 0$, use symmetric step 0.5.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/9

Slopes and local change

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Evaluate nearby points

J(0.5)=2.25 and J(-0.5)=6.25.



2. Estimate slope

Slope=(2.25-6.25)/(0.5-(-0.5))=-4.



3. Check small move

Move to a = 0.25: loss 3.0625, below starting loss 4.

CHANGE ONE THING

Move from 0 all the way to5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Evaluate nearby points	J(0.5)=2.25 and J(-0.5)=6.25.
2. Estimate slope	Slope=(2.25-6.25)/(0.5-(-0.5))=-4.
3. Check small move	Move to a = 0.25: loss 3.0625, below starting loss 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

What this example does not establish

Keep the stated limit

A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

Keep the stated limit

Read the labelled picture: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Move from 0 all the way to5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Slopes and local change claim and explain a repair using the supplied evidence.

A negative **slope** at w = 1 suggests that a sufficiently small increase reduces this smooth loss locally. Moving to 1.1 lowers loss from 1 to 0.81. A jump all the way to 4 raises loss to 4.

The initial **slope** cannot justify a step of any size.

A zero **slope** is also limited **evidence**. The nonnegative function F(w) = (w squared - 1) squared has zero local **slope** at w = 0, yet F(0) = 1 and F(0.1) = 0.9801. Small moves on either side lower that function.

A flat first-order clue can occur at a point that is not a minimum.

Locate the earliest unsupported step

Starting at w = 3, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Slopes and local change with support, then explain the deciding step in your own words.

For L(w) = (w - 2) squared, calculate the average **slope** from w = 0 to w = 2. Compare it with the local **slope** at w = 2.

At w = 3, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?

Explain your prediction to a partner before comparing results.

Practise with less help: Check a symmetric estimate

Use workbook W2 for the connected example "**Slope**s and **local change**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Slope**s and **local change**. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Slopes and local change. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

```
def loss(w):
    return (w - 2) ** 2

def local_slope(w, step):
    if step <= 0:
        raise ValueError("Step must be positive")
    return (loss(w + step) - loss(w - step)) / (2 * step)

for w in [1, 2, 3]:
    print(w, round(local_slope(w, 0.1), 6))
# Slopes near these centres: -2, 0, 2
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Slopes and local change. Explain why each step is justified.

Explain → apply → challenge

For L(w) = (w - 2) squared, calculate the average **slope** from w = 0 to w = 2. Compare it with the local **slope** at w = 2.

Starting at w = 3, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w.

> For F(w) = (w squared - 1) squared, compare F(0) and F(0.1). Does zero local **slope** at zero prove a minimum?

> **Explain which evidence supports your conclusion and which claim remains unproved.**

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **slope**s and **local change evidence** sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Always record the function, centre, nearby values and denominator used in a **slope** estimate. An average **slope** across a wide interval is not automatically the local **slope** at either endpoint.

Do not compare **slope** magnitudes without considering parameter and loss units.

A learner sees a negative estimated slope and says any larger parameter value will improve the loss. Give a counterexample using this lesson.

Next week changes one parameter at a time in a two-parameter model and combines those local **slope**s into a gradient.

For F(w) = (w squared - 1) squared, compare F(0) and F(0.1). Does zero local **slope** at zero prove a minimum?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Slopes and local change case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Compare slope clues on the same loss curve

CURRENT W	LOCAL SLOPE	NEARBY DOWNHILL DIRECTION
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Estimate a new curve independently

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Slopes and local change, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Use a local **slope** to propose a small move rather than claiming every move helps.

By the end, I can	How I will show it
Calculate an average rate of change over a stated interval	Explain the deciding step, show a trace or test, and state one limit.
Estimate a local slope with nearby loss values	Explain the deciding step, show a trace or test, and state one limit.
Explain the limits of a local direction clue	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Slopes and local change without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate an average rate of change over a stated interval

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use a local **slope** to propose a small move rather than claiming every move helps.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Move to a = 0.25: loss 3.0625, below starting loss 4.

1. Check small move

2. Estimate **slope**

3. Evaluate nearby points

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check small move Move to a = 0.25: loss 3.0625, below starting loss 4. A **slope** describes how the loss changes near the current parameter.

The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move from 0 all the way to5. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Loss becomes9, exceeding the starting4.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Loss becomes9, exceeding the starting4. A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Repair a missing building block

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

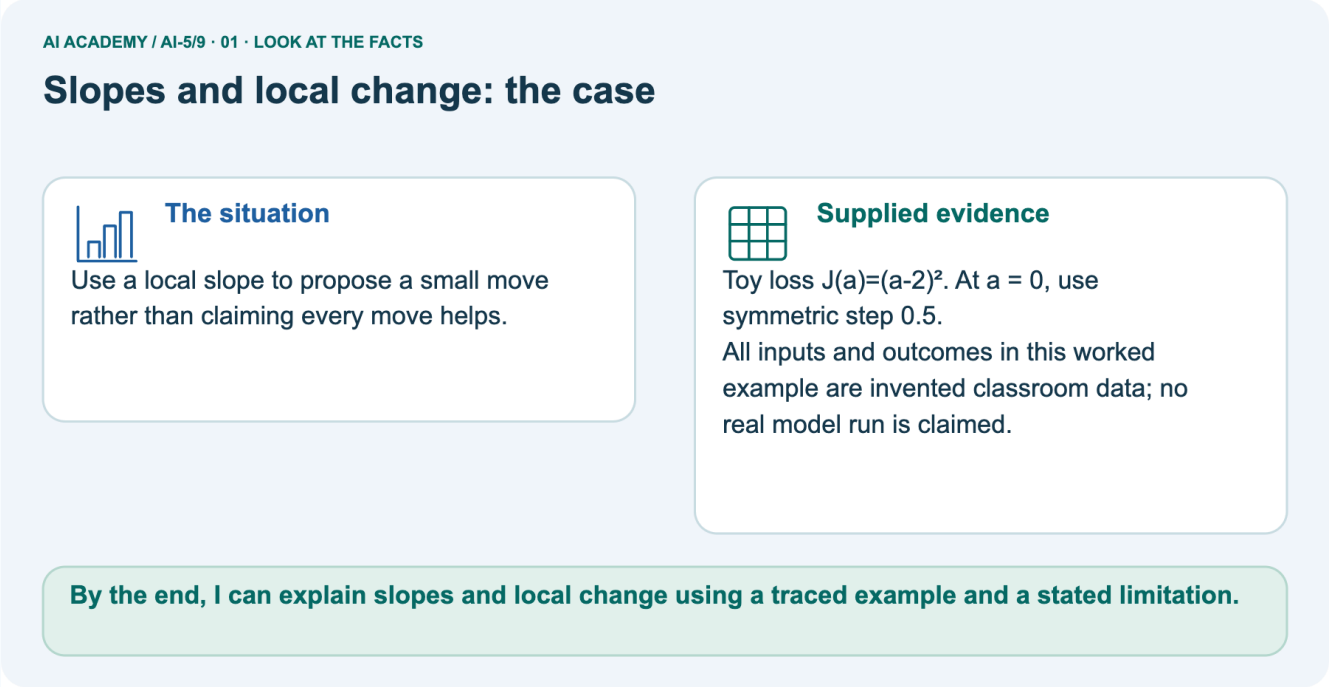
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Slopes and local change graphic using its labels, deciding evidence and relevant limitation.

01. Slopes and local change: the case



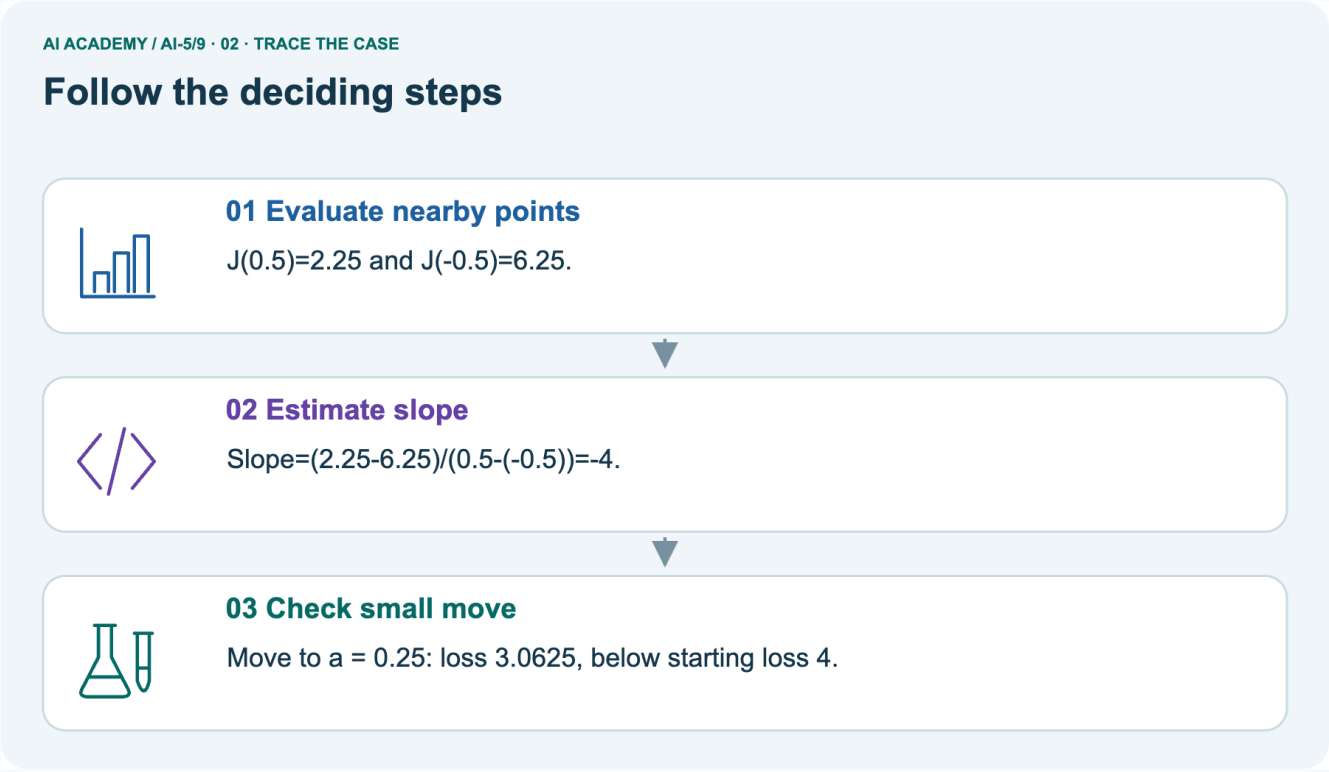
Follow the labelled picture with the matching explanation. Open the vector diagram

Use a local **slope** to propose a small move rather than claiming every move helps. Supplied facts: Toy loss J(a)=(a-2)². At a = 0, use symmetric step 0.5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Evaluate nearby points: J(0.5)=2.25 and J(-0.5)=6.25. → Estimate **slope**: **Slope**=(2.25-6.25)/(0.5-(-0.5))=-4. → Check small move: Move to a = 0.25: loss 3.0625, below starting loss 4.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Compare slope clues on the same loss curve

AI ACADEMY / AI-5/9 · 03 · INSPECT THE LESSON MODEL

Compare slope clues on the same loss curve

Current w	Local slope	Nearby downhill direction
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

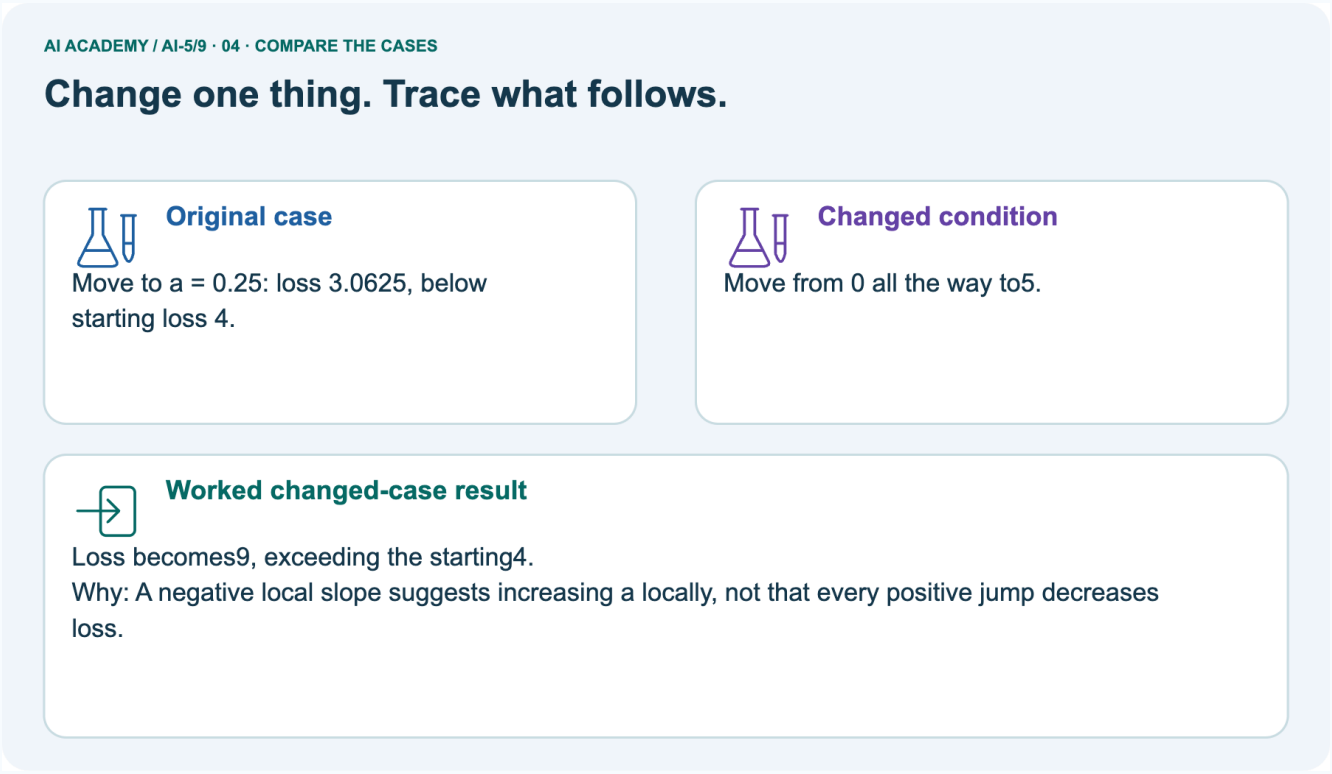
These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compare **slope** clues on the same loss curve. 1 | -2 | Increase w a little. 2 | 0 | No first order direction. 3 | 2 | Decrease w a little

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



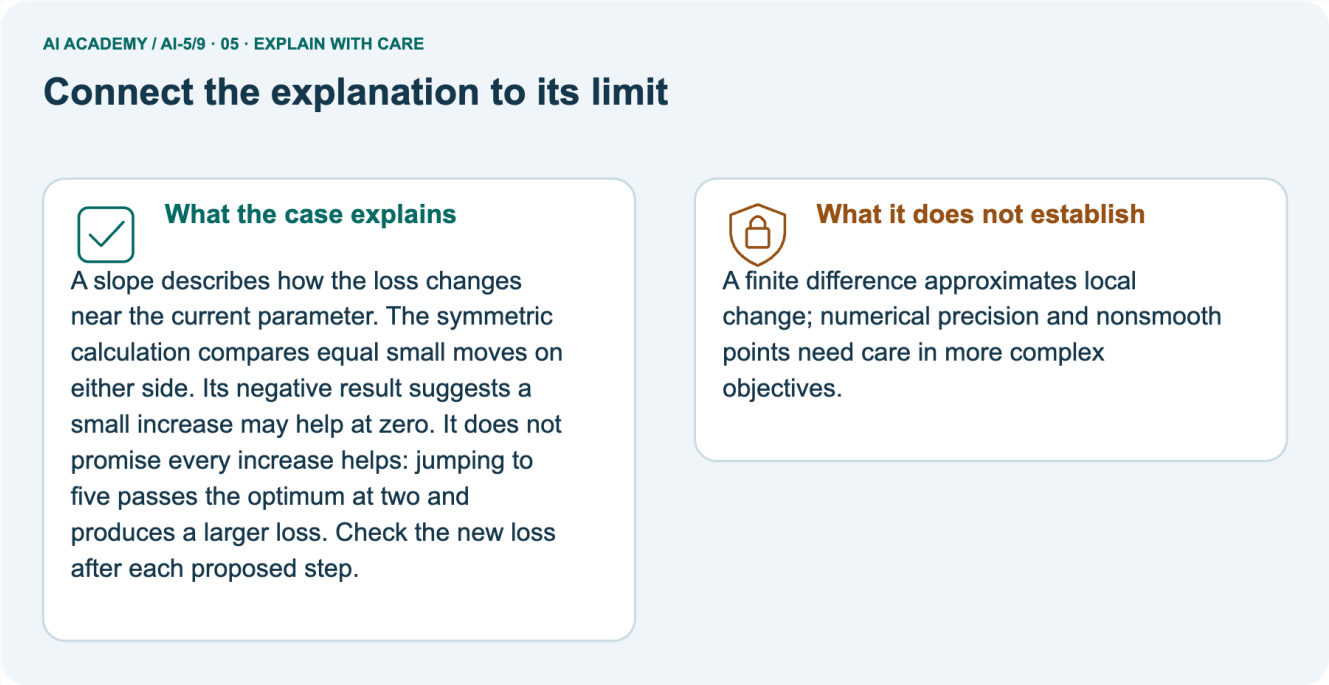
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Move to a = 0.25: loss 3.0625, below starting loss 4. Changed condition: Move from 0 all the way to5. Expected result: Loss becomes9, exceeding the starting4.

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

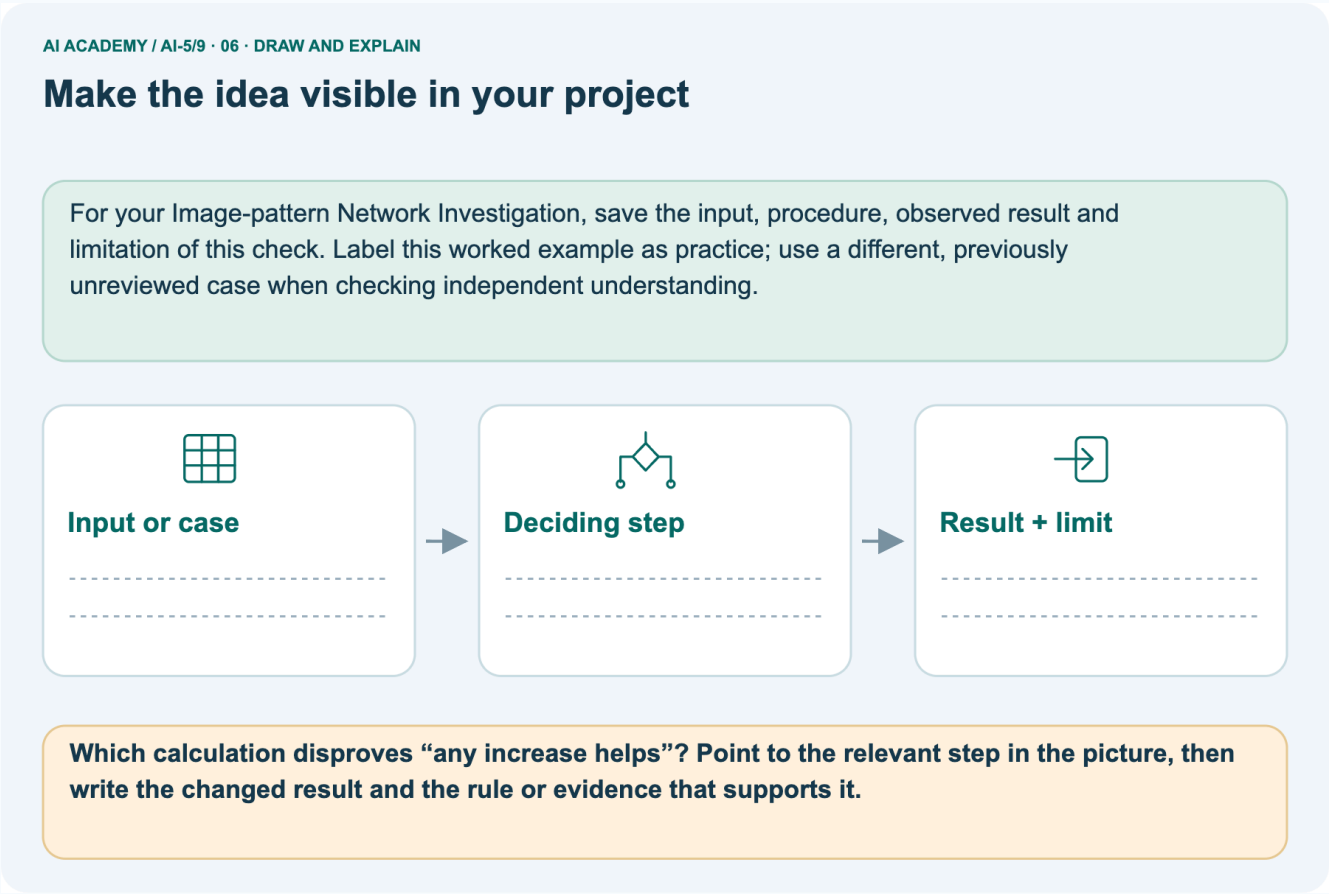
A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Limit: A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.